Lab Experiment – 12

Tic-Tac-Toe Game

Aim

To implement a Tic-Tac-Toe game using Python.

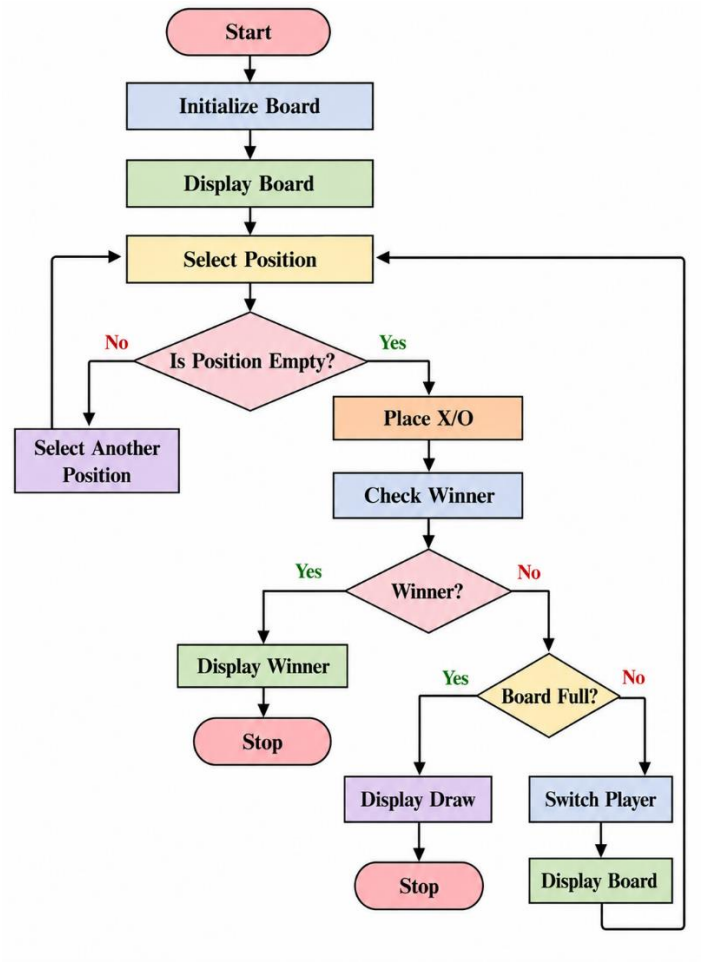
Objective

To develop a two-player Tic-Tac-Toe game where players take turns placing **X** and **O** on a 3×3 board.

Algorithm

1. Start the game.
2. Initialize a 3×3 board with empty spaces.
3. Display the board.
4. Ask the current player to select a position.
5. Place the player's symbol (X or O) in the selected position.
6. Check whether the player has won.
7. If a player wins, display the winner and stop.
8. Check whether the board is full.
9. If the board is full, declare the game as a draw.
10. Otherwise, switch to the other player.
11. Repeat the process until the game ends.

FlowChart



Code

```
def print_board(board):
    print("\n")
    print(" | ".join(board[0:3]))
    print("--+---+--")
    print(" | ".join(board[3:6]))
    print("--+---+--")
    print(" | ".join(board[6:9]))
    print()



def check_winner(board, player):
```

```
    winning_positions = [
        (0, 1, 2),
        (3, 4, 5),
        (6, 7, 8),
        (0, 3, 6),
        (1, 4, 7),
        (2, 5, 8),
        (0, 4, 8),
        (2, 4, 6)
    ]

    for a, b, c in winning_positions:
        if board[a] == board[b] == board[c] == player:
            return True

    return False



board = [" "] * 9
player = "X"

for turn in range(9):
    print_board(board)

    position = int(input(
        f"Player {player}, enter position (1-9): "
    )) - 1

    if position < 0 or position > 8 or board[position] != " ":
```

```
            print("Invalid position. Try again.")
            continue

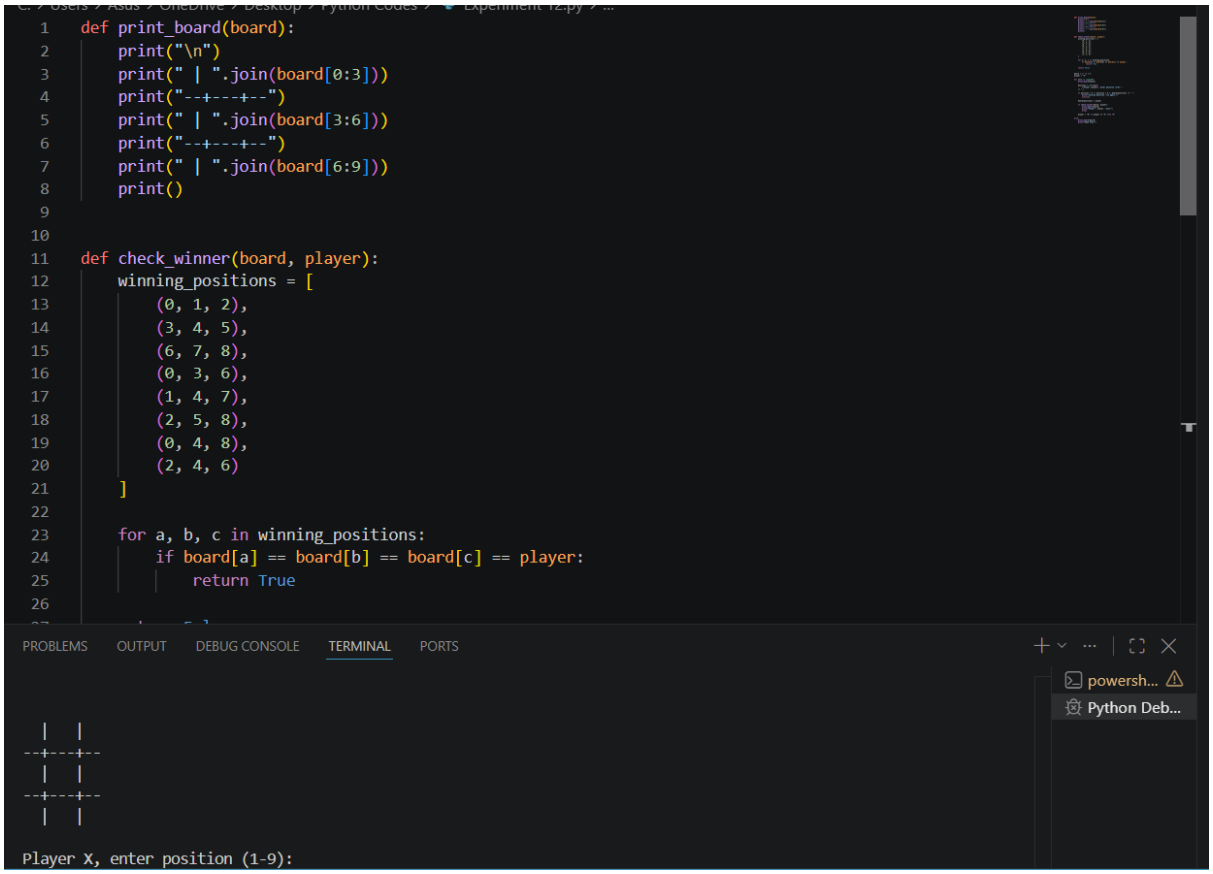
        board[position] = player

        if check_winner(board, player):
            print_board(board)
            print("Player", player, "wins!")
            break

        player = "O" if player == "X" else "X"

    else:
        print_board(board)
        print("Game Draw!")
```

Output



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        if board[a] == board[b] == board[c] == player:
            return True
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

  |   |
--+---+--
  |   |
--+---+--
  |   |

Player X, enter position (1-9):
```

Result

The Tic-Tac-Toe game was successfully implemented in Python.

Conclusion

The program successfully implements a two-player Tic-Tac-Toe game by managing player turns, validating positions, checking winning combinations, and identifying a draw when the board becomes full.